Chapter 4 | Contrasts



Learning objectives

- Convert a comparison of one-way ANOVA group means μ_i's to an expression of $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ for some vector $\mathbf{c}$.
- Convert a comparison of regression coefficient β_i's to an expression of $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ for some vector $\mathbf{c}$.
- Estimate a contrast $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ by $\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}$ and also compute the standard error of the contrast by $\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = \hat{\sigma}\sqrt{\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{c}}$.
- Construct a 95% confidence interval for a contrast $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$.
- Use a *t*-test to perform a hypothesis test of $\mathbf{c}^{\mathrm{T}}\boldsymbol{\beta} = 0$.

Outline

1. Introduction
2. Contrasts
3. Examples
4. Computation with R

ANOVA *F*-tests | Limitations

- If an ANOVA *F*-test rejects the null hypothesis $H_0 : \mu_1 = \cdots = \mu_m$, our conclusion is that not all the means are equal. That is, at least one mean is different from the other means.
- The ANOVA *F*-test itself provides only statistical evidence of a difference among means, but not any statistical evidence as to which mean or means are statistically different from others.
  1. In Example 2.2 and Lab 2-B, the ANOVA *F*-test resulted in a significant difference in mean phosphorous content between the three varieties with *p*-value = 0.0003.
  2. A follow up analysis is needed to determine which variety mean or means differ in phosphorous content.
  3. Plus we would want to know if one variety or multiple varieties were better/worse than another variety in average phosphorous content.

Cell-mean ANOVA | Limitations

- Consider a cell-mean specification of one-way ANOVA:

$$y_{ij} = \mu_i + \epsilon_{ij}, \qquad \epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$$

1. μ_i represents the unknown mean response of ith group (or treatment) for $i = 1, \ldots, m$.
2. The ANOVA model formula `lm( response ~ group + 0 )` computes $\hat{\mu}_i$'s and their associated standard errors, SE($\hat{\mu}_i$)'s, allowing us to compute CIs and perform a t-test for each mean $H_0 : \mu_i = 0$, but not for a mean difference.

```
> m_cell <- lm( phos ~ variety + 0, data=df.phos )   # "+0" indicates cell-mean
> summary( m_cell )

             Estimate Std. Error t value Pr(>|t|)
varietyvrt1   0.46000    0.04037   11.39 8.60e-08 ***
varietyvrt2   0.77600    0.04037   19.22 2.22e-10 ***
varietyvrt3   0.70800    0.04037   17.54 6.43e-10 ***
---
Residual standard error: 0.09028 on 12 degrees of freedom
```

Cell-mean ANOVA | Further comparison

- In the cell-mean ANOVA, how do we compute a 95% CI for $\mu_2 - \mu_3$?
  1. A 95% CI for $\mu_2 - \mu_3$ is
  $$\hat{\mu}_2 - \hat{\mu}_3 \pm t_{n-m,0.975}\text{SE}(\hat{\mu}_2 - \hat{\mu}_3)$$
  2. Note: $\text{SE}(\hat{\mu}_2 - \hat{\mu}_3) \neq \text{SE}(\hat{\mu}_2) - \text{SE}(\hat{\mu}_3)$, because
  $$\begin{aligned}
  \text{SE}(\hat{\mu}_2 - \hat{\mu}_3) &= \sqrt{\hat{\text{Var}}(\hat{\mu}_2 - \hat{\mu}_3)} \\
  &= \sqrt{\hat{\text{Var}}(\hat{\mu}_2) + \hat{\text{Var}}(\hat{\mu}_3) - 2\hat{\text{Cov}}(\hat{\mu}_2, \hat{\mu}_3)} \\
  &= \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^\top\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,3]}} \\
  &\neq \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,2]}} - \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[3,3]}} \\
  &= \text{SE}(\hat{\mu}_2) - \text{SE}(\hat{\mu}_3)
  \end{aligned}$$

Offset ANOVA | Limitations

- Consider an offset specification of one-way ANOVA:

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \qquad \epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$$

1. μ denotes the unknown mean response of 1st group (treatment) as a baseline.
2. τ_i is the difference of the unknown mean effect of ith group from 1st group, with $\tau_1 = 0$ in R.
3. Relation between cell-mean and offset: $\mu_1 = \mu$, $\mu_i = \mu + \tau_i$ for $i = 2, \ldots, m$.
4. The model formula `lm( response ~ group )` computes $\hat{\mu}$ and $\hat{\tau}_i$'s and their associated standard errors, SE($\hat{\mu}$) and SE($\hat{\tau}_i$)'s, allowing us to compute CIs and perform a t-test for each mean difference from the baseline $\mathrm{H}_0 : \tau_i = 0$, but not for a mean difference between others.

```
> m_offs <- lm( phos ~ variety, data=df.phos )
> summary( m_offs )

             Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.46000    0.04037  11.394  8.6e-08 ***
varietyvrt2   0.31600    0.05710   5.534 0.000129 ***
varietyvrt3   0.24800    0.05710   4.344 0.000956 ***
---
Residual standard error: 0.09028 on 12 degrees of freedom
```

Offset ANOVA | Further comparison

- In the offset ANOVA, how do we find a 95% CI for $\mu_2 - \mu_3 = (\mu + \tau_2) - (\mu + \tau_3) = \tau_2 - \tau_3$?
  1. A 95% CI for $\tau_2 - \tau_3$ is
  $$\hat{\tau}_2 - \hat{\tau}_3 \pm t_{n-m,0.975}\text{SE}(\hat{\tau}_2 - \hat{\tau}_3)$$
  2. Note: $\text{SE}(\hat{\tau}_2 - \hat{\tau}_3) \neq \text{SE}(\hat{\tau}_2) - \text{SE}(\hat{\tau}_3)$, because
  $$\begin{aligned}
  \text{SE}(\hat{\tau}_2 - \hat{\tau}_3) &= \sqrt{\hat{\text{Var}}(\hat{\tau}_2 - \hat{\tau}_3)} \\
  &= \sqrt{\hat{\text{Var}}(\hat{\tau}_2) + \hat{\text{Var}}(\hat{\tau}_3) - 2\hat{\text{Cov}}(\hat{\tau}_2, \hat{\tau}_3)} \\
  &= \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^\top\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,3]}} \\
  &\neq \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[2,2]}} - \hat{\sigma}\sqrt{(\mathbf{X}^\top\mathbf{X})^{-1}_{[3,3]}} \\
  &= \text{SE}(\hat{\tau}_2) - \text{SE}(\hat{\tau}_3)
  \end{aligned}$$
  3. Caution: $(\mathbf{X}^\top\mathbf{X})^{-1}$ is different for cell-mean and offset representations because $\mathbf{X}$ is different.

Contrasts for One-way ANOVA

- To complement the ANOVA F-test and t-test results, we use a method called **contrasts**.
- In the cell-mean ANOVA, $\mu_2 - \mu_3$ can be expressed as a linear function of $\boldsymbol{\beta} = [\mu_1, \mu_2, \mu_3]^{\mathrm{T}}$:

$$\mu_2 - \mu_3 = \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} \Rightarrow \boxed{\mu_2 - \mu_3 = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}} \text{ with } \mathbf{c} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \text{ and } \boldsymbol{\beta} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}$$

- In the offset ANOVA, $\mu_2 - \mu_3 = \tau_2 - \tau_3$ can be expressed as a linear function of $\boldsymbol{\beta} = [\mu, \tau_2, \tau_3]^{\mathrm{T}}$:

$$\tau_2 - \tau_3 = \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix} \Rightarrow \boxed{\tau_2 - \tau_3 = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}} \text{ with } \mathbf{c} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \text{ and } \boldsymbol{\beta} = \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix}$$

Contrasts | Foundational theory

- A **contrast** is a linear combination of elements of β that can be represented by the vector $\mathbf{c}$ such that the function we are interested in is estimated by $\mathbf{c}^T\hat{\beta}$.
- Recall: In the linear model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ with $\varepsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, LS estimators $\hat{\beta}$ has

$$E(\hat{\beta}) = \beta, \quad Var(\hat{\beta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$$

- Using some results of linear algebra, we have

$$E(\mathbf{c}^T\hat{\beta}) = \mathbf{c}^T\beta, \quad Var(\mathbf{c}^T\hat{\beta}) = \sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}$$

1. Why? $E(\mathbf{c}^T\hat{\beta}) = \mathbf{c}^T E(\hat{\beta}) = \mathbf{c}^T\beta$.
2. Also, $Var(\mathbf{c}^T\hat{\beta}) = \mathbf{c}^T Var(\hat{\beta})\mathbf{c} = \mathbf{c}^T\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c} = \sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}$.

- Standard error of $\mathbf{c}^T\hat{\beta}$:

$$SE(\mathbf{c}^T\hat{\beta}) = \sqrt{\hat{Var}(\mathbf{c}^T\hat{\beta})} = \hat{\sigma}\sqrt{\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}}$$

Example | Cell-mean ANOVA

- Example 4.1. Suppose a one-way ANOVA cell-mean specification with $m = 3$ groups (treatments). The mean difference between groups 2 and 3 can be expressed as

$$\mu_2 - \mu_3 = [0,\ 1,\ -1] \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$$

1. A 95% CI for $\mu_2 - \mu_3 = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ is

$$\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}} \pm t_{n-m,0.975}\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = [0,\ 1,\ -1] \begin{bmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \\ \hat{\mu}_3 \end{bmatrix} \pm t_{n-m,0.975}\,\hat{\sigma} \sqrt{[0,\ 1,\ -1](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}}$$

$$= \hat{\mu}_2 - \hat{\mu}_3 \pm t_{n-m,0.975}\,\hat{\sigma}\sqrt{(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,3]}}$$

2. A t-test statistic for $\mathrm{H}_0 : \mu_2 - \mu_3 = 0$ ($\mathrm{H}_0 : \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta} = 0$) is

$$t = \frac{\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}}{\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}})} = \frac{\hat{\mu}_2 - \hat{\mu}_3}{\hat{\sigma}\sqrt{(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,3]}}}$$

The p-value is $2P(t_{n-m} > |t|)$.

Example | Offset ANOVA

- Example 4.2. Consider a one-way ANOVA offset specification with $m = 3$ groups (treatments). The mean difference between groups 2 and 3 can be expressed as

$$\mu_2 - \mu_3 = (\mu + \tau_2) - (\mu + \tau_3) = \tau_2 - \tau_3 = [0,\ 1,\ -1] \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix} = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$$

1. A 95% CI for $\tau_2 - \tau_3 = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$ is

$$\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}} \pm t_{n-m,0.975}\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = [0,\ 1,\ -1] \begin{bmatrix} \hat{\mu} \\ \hat{\tau}_2 \\ \hat{\tau}_3 \end{bmatrix} \pm t_{n-m,0.975}\,\hat{\sigma} \sqrt{[0,\ 1,\ -1](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}}$$

$$= \hat{\tau}_2 - \hat{\tau}_3 \pm t_{n-m,0.975}\,\hat{\sigma} \sqrt{(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,3]}}$$

2. A t-test statistic for $\mathrm{H}_0 : \tau_2 - \tau_3 = 0$ ($\mathrm{H}_0 : \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta} = 0$) is

$$t = \frac{\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}}{\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}})} = \frac{\hat{\tau}_2 - \hat{\tau}_3}{\hat{\sigma}\sqrt{(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,2]} + (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[3,3]} - 2(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}_{[2,3]}}}$$

The p-value is $2P(t_{n-m} > |t|)$.

Example | SLR

- Example 4.3. From a SLR model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ for $i = 1, \ldots, n$, the LS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}$ is calculated from the data $(x_1, y_1), \ldots, (x_n, y_n)$. Now, suppose a new observation x_{new} of the predictor X is available.

  1. A 95% CI for the mean response at x_{new}, $\mathrm{E}(Y_{\text{new}}|x_{\text{new}}) = \beta_0 + \beta_1 x_{\text{new}} = [1,\ x_{\text{new}}] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$, is

$$\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}} \pm t_{n-2,0.975}\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{n-2,0.975}\,\hat{\sigma}\sqrt{[1,\ x_{\text{new}}](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\begin{bmatrix} 1 \\ x_{\text{new}} \end{bmatrix}}$$

  2. A 95% prediction interval (PI) for a new response y_{new} at x_{new}, $Y_{\text{new}} = \beta_0 + \beta_1 x_{\text{new}} + \epsilon_{\text{new}}$ which is estimated by $\hat{Y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}$, is

$$\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}} \pm t_{n-2,0.975}\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{n-2,0.975}\,\hat{\sigma}\sqrt{1 + [1,\ x_{\text{new}}](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\begin{bmatrix} 1 \\ x_{\text{new}} \end{bmatrix}}$$

  Note:

$$Var(\hat{Y}_{\text{new}}) = Var(\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} + \epsilon_{\text{new}}) = Var([1,\ x_{\text{new}}]\hat{\boldsymbol{\beta}}) + Var(\epsilon_{\text{new}}) = \sigma^2[1,\ x_{\text{new}}](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\begin{bmatrix} 1 \\ x_{\text{new}} \end{bmatrix} + \sigma^2$$

Contrast using `glht()`

- To perform statistical inference on $\mathbf{c}^T\boldsymbol{\beta}$, we need to specify the $m \times 1$[1] column vector $\mathbf{c}$.
  1. A 95% CI for $\mathbf{c}^T\boldsymbol{\beta}$ is
  $$\mathbf{c}^T\hat{\boldsymbol{\beta}} \pm t_{n-m,0.975}\text{SE}(\mathbf{c}^T\hat{\boldsymbol{\beta}}) = \mathbf{c}^T\hat{\boldsymbol{\beta}} \pm t_{n-m,0.975}\,\hat{\sigma}\sqrt{\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}}$$
  2. A t-test statistic for $H_0 : \mathbf{c}^T\boldsymbol{\beta} = 0$ is
  $$t = \frac{\mathbf{c}^T\hat{\boldsymbol{\beta}}}{\text{SE}(\mathbf{c}^T\hat{\boldsymbol{\beta}})} = \frac{\mathbf{c}^T\hat{\boldsymbol{\beta}}}{\hat{\sigma}\sqrt{\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}}}$$
  The p-value is $2P(t_{n-m} > |t|)$.
- We use the generalized linear hypothesis test function, `glht()`, from the `multcomp` package to do these computations.[2]

[1] For linear regression models with p predictors, replace $m \times 1$ with $p^* \times 1$, where $p^* = p + 1$.

[2] The p-values will be adjusted to correct for testing multiple hypotheses, so there may be slight differences compared to the p-value seen in the regular summary table.

Example | glht()

- Example 4.4. Refer to the phosphorus content data in Lab 2-B. We compare the mean phosphorus content of (1) variety 1 vs. variety 2 and (2) variety 2 and variety 3. That is, (1) $\mu_1 - \mu_2$ and (2) $\mu_2 - \mu_3$.

  1. Cell-mean: The cell-mean ANOVA model parameters are $\boldsymbol{\beta} = [\mu_1, \mu_2, \mu_3]^{\mathrm{T}}$. The contrasts are

$$\mu_1 - \mu_2 = [1, -1, 0] \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \mathbf{c}_1^{\mathrm{T}} \boldsymbol{\beta}, \quad \mu_2 - \mu_3 = [0, 1, -1] \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \mathbf{c}_2^{\mathrm{T}} \boldsymbol{\beta}$$

```
> m_cell <- lm( phos ~ variety + 0, data=df.phos )   # "+0" indicates cell-mean
> cntr_cell <- rbind( "Mean Diff: v1-v2" = c(1,-1, 0),
                      "Mean Diff: v2-v3" = c(0, 1,-1) )
> glht_cell <- multcomp::glht( m_cell, linfct=cntr_cell )
> summary( glht_cell )

                       Estimate Std. Error t value Pr(>|t|)
Mean Diff: v1-v2 == 0  -0.31600    0.05710  -5.534  < 0.001 ***
Mean Diff: v2-v3 == 0   0.06800    0.05710   1.191  0.62257
```

Because p-value = 0.6226 is greater than $\alpha = 0.05$, we do not reject $H_0 : \mu_2 - \mu_3 = 0$. There are no significance difference in the mean phosphorus content between varieties 2 and 3.

Example | glht()

- Example 4.4 (continued).
  2. Offset: The offset ANOVA model parameters are $\boldsymbol{\beta} = [\mu, \tau_2, \tau_3]^{\mathrm{T}}$. Note that

$$\mu_1 - \mu_2 = \mu - (\mu + \tau_2) = -\tau_2, \qquad \mu_2 - \mu_3 = (\mu + \tau_2) - (\mu + \tau_3) = \tau_2 - \tau_3$$

Therefore, the contrasts are

$$\mu_1 - \mu_2 = -\tau_2 = [0, -1, 0] \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix} = \mathbf{c}_1^{\mathrm{T}} \boldsymbol{\beta}, \quad \mu_2 - \mu_3 = \tau_2 - \tau_3 = [0, 1, -1] \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix} = \mathbf{c}_2^{\mathrm{T}} \boldsymbol{\beta}$$

```
> m_offs <- lm( phos ~ variety, data=df.phos )        # no "+0" indicates offset
> cntr_offs <- rbind( "Mean Diff: v1-v2" = c(0,-1, 0),
                      "Mean Diff: v2-v3" = c(0, 1,-1) )
> glht_offs <- multcomp::glht( m_offs, linfct=cntr_offs )
> summary( glht_offs )

                       Estimate Std. Error t value Pr(>|t|)
Mean Diff: v1-v2 == 0  -0.31600    0.05710  -5.534  < 0.001 ***
Mean Diff: v2-v3 == 0   0.06800    0.05710   1.191  0.62260
```

Because p-value = 0.6226 is greater than $\alpha = 0.05$, we do not reject $\mathrm{H}_0 : \mu_2 - \mu_3 = 0$. There are no significance difference in the mean phosphorus content between varieties 2 and 3.

Contrast using `emmeans()`

- Specifying the contrasts by hand can be laborious and difficult to do correctly.
- The `emmeans()` function in the `emmeans` package bypasses the creation of **c** by considering all possible pairwise contrasts.

Example | emmeans()

- Example 4.5. Refer to the phosphorus content data in Lab 2-B. With three varieties, there are a total of 3 possible pairwise comparisons: (1) $\mu_1 - \mu_2$, (2) $\mu_1 - \mu_3$, (3) $\mu_2 - \mu_3$.

1. Cell-mean ANOVA

```
> emmeans_cell <- emmeans( m_cell, specs=pairwise~variety, infer=c(TRUE,TRUE) )
> summary( emmeans_cell )

   contrast estimate     SE  df  lower.CL  upper.CL  t.ratio   p.value
vrt1 - vrt2   -0.316  0.057  12   -0.4683   -0.1636   -5.534   0.00035
vrt1 - vrt3   -0.248  0.057  12   -0.4003   -0.0956   -4.343   0.00254
vrt2 - vrt3    0.068  0.057  12   -0.0843    0.2203    1.191   0.48047
```

   With 95% confidence, we estimate the difference in the mean phosphorus content between varieties 2 and 3 is $(-0.0843, 0.2203)$.

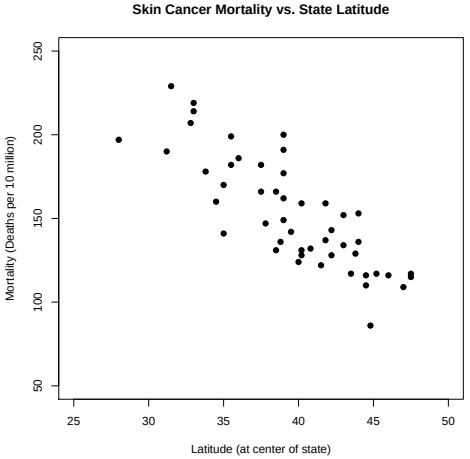
2. `emmeans()` produces the same result for offset ANOVA.

```
> emmeans_offs <- emmeans( m_offs, specs=pairwise~variety, infer=c(TRUE,TRUE) )
> summary( emmeans_offs )
```

Example | SLR

- Example 4.6. The response variable Y is the mortality due to skin cancer (number of deaths per 10 million people) and the predictor variable X is the latitude (degrees North) at the center of each of 49 states in the U.S.[3]

	State	Lat	Mort	Ocean	Long
1	Alabama	33.0	219	1	87.0
2	Arizona	34.5	160	0	112.0
3	Arkansas	35.0	170	0	92.5
4	California	37.5	182	1	119.5
5	Colorado	39.0	149	0	105.5
6	Connecticut	41.8	159	1	72.8
7	Delaware	39.0	200	1	75.5
8	Wash,D.C.	39.0	177	0	77.0
:	:	:	:	:	:
45	Virginia	37.5	166	1	78.5
46	Washington	47.5	117	1	121.0
47	WestVirginia	38.8	136	0	80.8
48	Wisconsin	44.5	110	0	90.2
49	Wyoming	43.0	134	0	107.5



[3] Dataset: `skin_cancer.txt`. The data were compiled in the 1950s, so Alaska and Hawaii were not yet states. And, Washington, D.C. is included even though it is not technically a state.

Example | SLR

- Example 4.6 (Q1). We estimate the mean mortality at a location with latitude = 30 using 95% CI.

1. A 95% CI for the mean response at x_{new}, $\mathrm{E}(Y_{\text{new}}|x_{\text{new}}) = \beta_0 + \beta_1 x_{\text{new}} = [1,\ x_{\text{new}}] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \mathbf{c}^{\mathrm{T}}\boldsymbol{\beta}$, is

$$\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}} \pm t_{n-2,0.975}\mathrm{SE}(\mathbf{c}^{\mathrm{T}}\hat{\boldsymbol{\beta}}) = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{n-2,0.975}\,\hat{\sigma}\sqrt{[1,\ x_{\text{new}}](\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\begin{bmatrix} 1 \\ x_{\text{new}} \end{bmatrix}}$$

2. Use `predict()` to compute CIs.

```
> m_slr <- lm( Mort~Lat, data=df.skin )
> df.skin_new <- data.frame( Lat=30 )
> predict( m_slr, newdata=df.skin_new, level=0.95, interval="confidence" )
       fit         lwr         upr
209.8603   197.1380   222.5826
```

3. We also use `emmeans()`.

```
> summary( emmeans( m_slr, specs=~Lat, at=list(Lat=30), infer=c(TRUE,TRUE) ) )
Lat    emmean       SE  df  lower.CL  upper.CL  t.ratio   p.value
 30  209.8603  6.3240  47  197.1380  222.5826  33.1846  2.85e-34
```

Example | SLR

- Example 4.6 (Q2). An advantage of using `emmeans()` is its ability to apply to many different pairwise comparisons.
  1. Consider the difference in the mean mortality between a location at latitude 40 and a location with latitude 30. That is,

$$E(Y|X=40) - E(Y|X=30) = (\beta_0 + \beta_1 \cdot 40) - (\beta_0 + \beta_1 \cdot 30) = 10\beta_1 = [0, 10] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \mathbf{c}^{\mathrm{T}} \boldsymbol{\beta}$$

  2. "Hand calculation" by matrix operation will use $\mathbf{c}^{\mathrm{T}} = [0, 10]$.
  3. We can implement this contrast using `emmeans()`.

```
> summary( emmeans( m_slr, specs=revpairwise~Lat, at=list( Lat=c(30,40) ),
                    infer=c(TRUE,TRUE) ) )
     contrast  estimate      SE  df  lower.CL  upper.CL  t.ratio  p.value
Lat40 - Lat30  -59.7763  5.9837  47  -71.8140  -47.7386  -9.9898  3.3e-13
```

Because p-value is almost 0, we reject $H_0 : E(Y|X=40) - E(Y|X=30) = 0$. There are significant difference in the mean mortality between locations with latitudes 40 and 30.